PANJAB UNIVERSITY, CHANDIGARH

(Estted. under the Panjab University Act VII of 1947- enacted by the Govt. of India)

FACULTY OF ARTS

SYLLABI

FOR

MASTERS IN REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS (GIS) (SEMESTER SYSTEM)

EXAMINATIONS, 2011-2012

PANJAB UNIVERSITY, CHANDIGARH

OUTLINES OF TESTS, SYLLABI AND COURSES OF READING FOR MASTERS IN REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS (GIS) (SEMESTER SYSTEM) FOR THE EXAMINATIONS OF 2011 -2012.

COURSE STRUCTURE

SEMESTER I

		Max. Marks
Paper I:	Fundamentals of Remote Sensing	**200**
	(a) Theory : 100	
	(b) Practicals : 60	
	(Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)	
	(c) Internal Assessment : 40	
Paper II:	Fundamentals of Geographic Information Systems (GIS)	**200**
	(a) Theory : 100	
	(b) Practicals : 60	
	(Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)	
	(c) Internal Assessment : 40	
Paper III:	Spatial Analysis	**200**
	(a) Theory : 100	
	(b) Practicals : 60	
	(Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)	
	(c) Internal Assessment : 40	

SEMESTER II

Paper IV:	Remote Sensing and GIS in Human geography (with special reference to Urban and Regional Planning)	**200**
	(a) Theory : 100	
	(b) Practicals : 60	
	(Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)	
	(c) Internal Assessment : 40	
Paper V:	Remote Sensing and GIS in Physical Geography (with special reference to environmental management)	**200**
	(a) Theory : 100	
	(b) Practicals : 60	
	(Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)	
	(c) Internal Assessment : 40	
Paper VI:	Image Processing	**100**
	(a) Theory : 50	
	(b) Practicals : 30	
	(Lab. work 15 marks+ Record file 10 marks+ Viva 5 marks)	
	(c) Internal Assessment : 20	
	Field Report	**100**
	(a) Evaluation of the Field Report : 50	
	(b) Presentation : 25	
	(c) Viva Voce : 25	

SEMESTER III

Paper VII:	Application of Remote Sensing and GIS in Disaster Management		**200**
	(a) Theory	: 100	
	(b) Practicals	: 60	
	(Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)		
	(c) Internal Assessment	: 40	
Paper VIII:	Application of Remote Sensing and GIS in Utility Management		**200**
	(a) Theory	: 100	
	(b) Practicals	: 60	
	(Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)		
	(c) Internal Assessment	: 40	
Paper IX:	Application of Remote Sensing and GIS in Resource Management		**200**
	(a) Theory	: 100	
	(b) Practicals	: 60	
	(Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks)		
	(c) Internal Assessment	: 40	

SEMESTER –IV

Paper X:	Research Methodology, Project Formulation and Writing		**100**
	(a)Theory	: 80	
	(b) Internal Assessment	: 20	
	Project Report		**500**
	(a) Mid-Term Appraisal of the Project Report	: 100	
	(b) Evaluation of the Project Report	: 325	
	(c) Presentation &Viva Voce (50+25)	: 75	
	Grand Total		**2400**

NOTE:

1. The candidates, who successfully completes first two semesters of the course but scores less than 60 per cent marks in aggregate, shall be awarded Post-Graduate Diploma in Remote Sensing and Geographic Information Systems (GIS).
2. Admission to the third semester will be restricted to those students who have secured at least 60 per cent marks in the first two semesters of this Course.
3. Candidates, who have already done one-year Post-graduate Diploma in Remote Sensing and Geographic Information Systems from this department and wish to do Masters in Remote Sensing and Geographic Information Systems (GIS) shall be admitted to the third semester provided their score of the said Post-graduate Diploma falls among the top 25 per cent of the admitted candidates of the on going Masters in Remote Sensing and Geographic Information Systems (GIS) class, subject to availability of seats.
4. The minimum marks required to pass the examination shall be 45% in each (i) theory paper, (ii) practical work, and (iii) field/project report, and 50% in aggregate.
5. The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.
6. No candidate shall be allowed to do Masters in Remote Sensing and Geographic Information Systems (GIS) after a gap of three years of passing of P.G. Diploma in Remote Sensing and Geographic Information Systems (GIS)

SEMESTER – I

PAPER - I: FUNDAMENTALS OF REMOTE SENSING **Marks: 200**

(A) THEORY **Marks: 100**
Time : 3 Hours

Objectives:

- to introduce the students about the basic principles of remote sensing;
- to indicate the methods of visual and digital interpretation of aerial photographs and satellite imageries;
- to explain the methods of acquiring remote sensing data and their usage; and
- to train the students about elementary photogrammetry and use of GPS.

Course contents:

Unit-I

Introduction to remote sensing: Energy and radiation principles, energy-atmosphere interaction, energy-earth surface features interaction, spectral signatures.
Resolution of Remote Sensing data: Spatial, spectral, radiometric and temporal.

Unit-II

Remote sensing types: Natural, technology assisted, active, passive, photographic, non-photographic, etc.
Type of aerial photographs and satellite imageries: Classification, characteristics and applications.
Sensor Platforms: Ground based, air borne and space borne, earth resources satellite systems: LANDSAT, SPOT, IRS, ERS, IKONOS, etc.

Unit-III

Image errors due to tilt, relief, optical distortions, image restoration and rectification.
Photogrammetry: Simple geometry of vertical photograph measurement of scale, height, and slope from the vertical photographs.

Unit-IV

Digital image processing: Introduction to image processing system: hardware and software.
GPS and its use in the selection of training sets and field verification of interpreted data.

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit, and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(B) PRACTICAL **Marks: 60**

(i) Arrangement of aerial photographs with the help of numbers given on the photograph and preparation of photo index.

(ii) Interpretation of information given on the margins of the photographs and satellite imageries.

(iii) Stereo vision test, selection of stereopair, and its orientation and placing under the stereoscope for stereovision.

(iv) Identification and mapping of features from a single photograph

(v) Identification and mapping of features from a stereopair and their verification in the field.

(vi) Locating principal point, conjugate principal point and flight line on a stereopair.

(vii) Measurement of scale, relief displacement, height and slope from vertical photographs.

(viii) Removing error due to relief displacement on vertical photographs by graphical method.

(ix) Display of digital data, geometric and radiometric correction.

(x) Identification of spectral signatures of some selected features and histogram analysis.

(xi) Image rectification, restoration and enhancement exercises using different methods.

(xii) Supervised and unsupervised classification of digital images.

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT: Marks: 40

Internal Assessment shall be based on class test/s and assignment/s etc., as prescribed or conducted by the teacher/s concerned.

BOOKS RECOMMENDED

ESSENTIAL READINGS:

American Society of Photogrammetry : *Manual of Remote Sensing*, Vol. 1 and II, Falls Church, Virginia, 1975.

Barret, E.C. & Curtis, E.F. : *Remote Sensing of Environment*, Second Edition, 1982.

Curran, Paul J. : *Principles of Remote Sensing*, Longman, Hongkong, 1988.

Jensen, J.R. : *Introductory Digital Image Processing: A Remote Sensing Perspective*, Prentice Hall, New Jersey, 1986.

Lillesand, T.M. & Kiefer, R.W. : *Remote Sensing and Image Interpretation*, Third Edition, John Wiley and Sons, 1994.

FURTHER READINGS:

Agarwal, C.S. and P.K. Garg : *Text Book on Remote Sensing in Natural Resources Monitoring and Management*, Wheeler Publishing Co, New Delhi, 2000.

Jain, A.K. : *Fundamentals of Digital Image Processing*, Prentice Hall, New Jersey, 1989.

Muralikrishna, I.V. : *Remote Sensing Applications and Geographic Information Systems Recent Trends*, Tata McGraw Hill, New Delhi, 1992.

Fratt, W.K. : *Digital Image Processing*, John Wiley & Sons, 1978.

Rashid, S.K. : *Remote Sensing in Geography*, Manak Publications, Delhi, 1993.

Rees, W.G. : *Physical Principles of Remote Sensing*, Cambridge University Press, Cambridge, 1990.

Richards, John, A. : *Remote Sensing*: Digital Image Analysis, Springer Verlag, 1986.

Swain, Phillip, H. : *Remote Sensing*: The quantitative Approach, McGraw Hill International Book Co., 1978.

Verbyla, David, L : *Satellite Remote Sensing of Natural Resources*, Lewis Publishers, New York, 2005

PEDAGOGY:

- Students may be taken to field for verification of identifiable features on aerial photographs and satellite imageries;
- Students may be taken to different institutions, such as IIRS, NRSA, State Remote Sensing Centres to acquaint with different equipments, techniques, and their products; and
- Students may be asked to prepare a report using toposheets, aerial photographs and satellite imageries to trace the temporal and spatial changes.

PAPER II: FUNDAMENTALS OF GEOGRAPHIC INFORMATION SYSTEMS (GIS) **Marks: 200**

(A) THEORY **Marks: 100**
Time : 3 Hours

OBJECTIVES:
The main objective of this course is to expose the students to fundamental principles of Geographical Information Systems including basic concepts and definitions, methods and techniques, and mathematical models.

COURSE CONTENTS:

Unit-I

Concepts and Definitions:
- Geographic Information Systems (GIS): Definition and developments
- Spatial data base: Points, Lines and Polygons
- Computer components: Hardware and software

Unit-II

Functional Elements:
- Data acquisition, input, editing, data manipulation and topology creation
- Data analysis and query
- Symbolisation and product generation

Unit-III

Data Management and Structure:
- Data Base Management System (DBMS): Purpose and structure
- Classification: Raster, Vector, quadtree, hierarchical, network and relational

Unit-IV

Artificial Intelligence and Neural Networks:
- Artificial intelligence: history, major issues, approaches and pattern recognition.
- Artificial neural network: Definition, taxonomy and multi-layer perceptions

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit, and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(B) PRACTICAL **Marks: 60**

(i) Windows/ NT Environment: Basics, file and directory organisation and management.

(ii) Computer data-binary notation, ASC-II coding system

(iii) Boolean operations

(iv) GIS database: Network, hierarchical, and relational database, creating raster and vector GIS data; creating buffer zones; overlay and using raster and vector GIS.

(v) Using PC Arc/Info or other GIS software: Digitizing, editing, map displaying, query.

(vi) Elevation Representation: Point, line, triangular.

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 40**

Internal Assessment shall be based on class test/s and assignment/s etc., as prescribed or conducted by the teacher/s concerned.

BOOKS RECOMMENDED

ESSENTIAL READINGS

Aronoff, S. : *Geographic Information Systems: A Management Perspective*, WDL Publications Ottawa,Canada, 1992.

ESRI : *Understanding GIS*, Environmental Systems Research Institute, U.S.A., 1993.

Jefrey S. & John, E. : *Geographic Information Systems - An Introduction*, Prentice Hall, New Jersey, USA, 1990.

Michael F. Goodchild & Karan K. Kemp (eds.) : *Introduction to GIS*, Santa Barbara, California, NGGIA, 1990.

FURTHER READINGS

Elaine, Rich : *Artificial Intelligence*, McGraw Hill Book Co., 1983.

Giarratano, J. & Rieley, G. : *Expert Systems: Principles and Programming*,PRS Kent Publishing Co., 1985.

Laurini, R. & Thomson, D. : *Fundamentals of Spatial Information Systems*, Academic Press, London, 1994.

Rashid, S.M. : *Remote Sensing in Geography*, Manak Publications, Delhi, 1993.

PEDAGOGY:

Distribution of a brief synopsis of different topics prior to discussions in the class, involving students in teaching and arranging question-answer session at the end will form the core of class-teaching approach. Use of audio-visual aids such as transparencies, slides and power point presentations will be guiding principles of teaching methodology. Organising occasional field visits and inviting professionals as a guest faculty will be used to differentiate between the theory and practice.

PAPER -III: SPATIAL ANALYSIS **Marks: 200**

(A) THEORY **Marks: 100**
Time : 3 Hours

OBJECTIVE:

The course is designed to help the students to understand the techniques, which assist in providing focused and precise analysis of emerged pattern of geographical phenomena.

COURSE CONTENT:

Unit- I

Introduction to Spatial analysis:

- Fundamentals of Spatial analysis: Scales of measurement; Scale, extent and Projection
- Significance of Spatial analysis.

Unit- II

Point Pattern Analysis

- The meaning and nature of Point features over space
- Measuring Central tendency
  - Mean Centre, Weighted Mean Centre, Median Centre
- Dispersion of Point features
  - Standard distance, Standard Deviation Ellipse
- Quadrat Analysis, Nearest Neighbour statistics
- Spatial Autocorrelation of Point features
  - Geary's Ratio
  - Moran's I Index

Unit –III

Line Pattern Analysis

- The meaning, nature and characteristics of Line features over space
- Directional Mean and Circular Variance
- Network Analysis
  - Connectivity or Topology
  - Proximity Analysis

Polygon Pattern Analysis

- The meaning of spatial relationships and spatial dependency
- Spatial Autocorrelation
  - Geary's Ratio
  - Moran's I Index

Bivariate Spatial Autocorrelation

Unit-IV

Vector Analysis

- Buffering: Point, Line, Area buffers
- Overlay
  - Union, Intersect, Symmetrical Difference, Identity
  - Point-in-Polygon, Line-in-Polygon, Polygon-in-Polygon

Surface Analysis

- Digital Elevation Model (DEM) and Digital Terrain Model (DTM)
- Contouring, Hill Shading and Perspective View (3D viewing)
- Slope and Aspect Analysis

Note:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt all 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit.
3. The candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(B) PRACTICAL **Marks: 60**

The examination in 'practicals' shall be conducted by a team of three examiners, including an internal, the Chairperson of the Department, and one additional faculty member (as appointed by the Board of Control)

(i) Familiarization with Spatial Analysis Software.

(ii) Spatial analysis: Query, Identifying, measuring, query by location/attribute

(iii) Measuring Central tendency; Mean Centre, Weighted Mean Centre, Median Centre, Standard distance, Standard Deviation Ellipse

(iv) Quadrat Analysis, Nearest Neighbour statistics

(v) Spatial Autocorrelation of Point features; Geary's Ratio, Moran's I Index

(vi) Network analysis

(vii) Vector Analysis; Buffering: Point, Line, Area buffers

(viii) Overlay

(ix) Surface Analysis; Digital Elevation Model (DEM) and Digital Terrain Model (DTM), Contouring, Hill Shading and Perspective View (3D viewing), Slope and Aspect Analysis.

(C) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be based on class test/s and assignment/s etc., as prescribed or conducted by the teacher/s concerned

BOOKS RECOMMENDED

ESSENTIAL READINGS

Aronoff, S. : *Geographic Information Systems: A Management Perspective***,** WDL Publications Ottawa, Canada, 1992.

ESRI : *Understanding GIS*, Environmental Systems Research Institute, U.S.A., 1993.

Jefrey S. & John, E. : *Geographic Information Systems - An Introduction.* Prentice Hall, New Jersey, USA, 1990.

Michael F. Goodchild & Karan K. Kemp (eds.) : *Introduction to GIS*, Santa Barbara, California, NGGIA, 1990.

Wong, David W.S. & Jay Lee: Statistical Analysis of Geographic Information with ArcView GIS and ArcGIS, John Wiley and Sons, INC., New Jersey, USA, 2005.

FURTHER READINGS

Agarwal, C.S. and P.K. Garg : Text Book on Remote Sensing in Natural Resources Monitoring and Management, Wheeler Publishing Co., New Delhi, 2000.

Burrough, P.A. & Macdoneli, R.A. : *Principles of Geographic Information Systems,* Oxford University Press, 2000.

Demers, M.N. : *Fundamentals of geographic Information Systems*, 2nd Edition, John Wiley and sons, 2000.

Fratt, W.K. : *Digital Image Processing, John Wiley & Sons, 1978.*

Laurini, R. & Thomson, D. : *Fundamentals of Spatial Information Systems*, Academic Press, London, 1994.

Muralikrishna, I.V. : Remote Sensing Applications and Geographic Information Systems Recent Trends, Tata McGraw Hill, New Delhi, 1992.

Rashid, S.K. : Remote Sensing in Geography, Manak Publications, Delhi, 1993.

Rees, W.G. : Physical Principles of Remote Sensing, Cambridge University Press, Cambridge, 1990.

Richards, John, A. : Remote Sensing: Digital Image Analysis, Springer Verlag, 1986.

Swain, Phillip, H. : Remote Sensing: The Quantitative Approach, McGraw Hill International Book Co., 1978.

Verbyla, David, L : Satellite Remote Sensing of Natural Resources, Lewis Publishers, New York, 2005.

PEDAGOGY:

Students will be encouraged to ask questions and participate in classroom discussions. Assignments, term papers and group discussion will be the additional methods to make teaching more focussed and interesting.

SEMESTER II

PAPER IV: REMOTE SENSING & GIS IN HUMAN GEOGRAPHY **Marks: 200**
(with special reference to Urban and Regional Planning)

(A) THEORY **Marks: 100**
Time : 3 Hours

OBJECTIVES:

To provide an understanding of:

- concepts and definitions of urban and regional planning so as to pin point the specific areas, where application of Remote Sensing and GIS technologies will increase efficiency and effectiveness of planning;
- data base creation in terms of various attributes, types and hierarchical scales; and
- planning techniques which help in the location of infrastructural facilities for decentralised urban and regional planning.

COURSE CONTENT:

Unit-I

Concepts and definitions: Planning - definition and types; role of remote sensing and GIS in regional planning, tourism and business management.
Regional Planning: Creation of data base for regional planning; land use/ land cover mapping; land use suitability evaluation; updating of land use/ land cover maps; resource analysis for district/ block planning; location of amenities and facilities.

Unit-II

Creation of data base for urban planning; updating of urban land use maps; classification of residential areas, estimation of population concentrations, urban growth monitoring, and urban change detection; route location and site planning, urban development planning; location of facilities and creation of data base for urban planning; slum identification and improvement.

Unit-III

Urban and regional mapping, Base map preparation, regional, city, intra–city, scale, methodology.
Urban and regional plans formulation, Application of remote sensing techniques, Regional plan. master plan, detailed development plan- Methodologies and stages – Case studies, coastal and wasteland development plans.
Urban analysis, Urban growth, trend analysis, change detection, slum development, housing typology and density analysis population estimation, environmental quality rating- Transportation network analysis- Case studies.

Unit-IV

Information system, Database Organisation- Geographic Information System- Large scale data entry- Interpretation manipulation- Retrieval- Suitable software Package- Attribute information for urban planning
Urban analysis and modeling with GIS, Decision support system for urban and regional management.

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(B) PRACTICAL **Marks: 60**

(i) Image/photo interpretation: recognition and detection of objects, preparation of land use maps.

(ii) Digital image processing techniques in preparation of land use and urban land use maps.

(iii) Urban growth monitoring and change detection.

(iv) Preparation of base map and its updating.

(v) Grid cell data processing.

(vi) Site planning of business activities and tourist spots.

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be as based on class test/s and assignment/s etc., as prescribed or conducted by the teacher/s concerned.

BOOKS RECOMMENDED

ESSENTIAL READINGS

Avery, T.V. : *Interpretation of Aerial Photography,* Burgess Publishing Co., Minneapolis, 1962.

Barret, E.C. & Curtis, E.F. : 'The Built Environment' in Remote Sensing of Environment, Second edition, pp. 293-306, 1982.

Gautam, N.C. : *Urban Landuse Study through Aerial Photo Interpretation Techniques*, Pink Publishing House, Mathura, 1970.

Muralikrishna, I.V. : *Remote Sensing Applications and Geographic Information Systems - Recent Trends*, Tata McGraw Hill, New Delhi, 1992.

Nag, Prithvish : Thematic Cartography and Remote Sensing, Concept, New Delhi, 1992.

Richardson, B.F(ed) : Introduction to Remote Sensing of the Environment, Kendall/Hunt, Dubuque, Iowa, 1978.

FURTHER READINGS

Sundaram, K.V. : Urban and Regional Planning in India, Concept, New Delhi, 1977.

Taylor, John,L. Williams, David C : Urban Planning Practice in Developing Countries, Pergamon Press, 1981.

PEDAGOGY:

Taking students for fieldwork, inviting professionals for lectures, taking students to institutes/training centres engaged in imparting professional training and providing research consultancy in business, tourism and area/regional planning should be the main focus.

PAPER V: REMOTE SENSING AND GIS IN PHYSICAL GEOGRAPHY **Marks: 200**
(with special reference to Environmental Management)

(A) THEORY **Marks: 100**
Time : 3 Hours

OBJECTIVES:

- to train the students to detect, identify and recognise the various facets of different landforms using remote sensing and creating a data base in GIS mode;
- to educate students about morphogenetic analysis using photogrammetry & GIS; and
- to indicate the use of remote sensing and GIS in assessment and management of environmental problems.

COURSE CONTENT:

Unit –I

Concepts and definitions:

- Basic concepts in geomorphology, climatology, oceanography and biogeography
- Environment: Definition and components
- Human-environment relations

Unit –II

Remote Sensing and GIS in Geomorphology:

- Terrain evaluation and mapping of terrain units at various scales using remote sensing and GIS
- Manual and digital methods of slope analysis.
- Manual and digital methods of morphometric analysis of a drainage basin.
- Use of remote sensing in understanding the relationships between structure and process of landforms, climate and landforms structure and drainage pattern, landforms and surface material, landforms and landuse, landforms and human activities.
- Classification of landforms using remote sensing and GIS

Unit-III

Remote Sensing and GIS in climatology, hydrology and environmental geomorphology and management:

- Earth/ atmosphere energy and radiation budget.
- Use of remote sensing and GIS in recording the meteorological data and weather forecasting.
- Use of remote sensing and GIS in hydrological studies, especially mapping of water resources and water related problems.
- The concept of environmental geomorphology
- Use of Remote Sensing & GIS in impact assessment of human activities on the geomorphological processes and the landscape
- Fundamentals and goals of environmental management
- Strategies for environmental management

Unit –IV

- Use of Remote Sensing and GIS in assessment of environmental problems such as deforestation, soil erosion, eutrophication, water logging soil salinity/alkalinity, desertification, waste generation, spread of weeds, water, air and soil pollution etc.

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(B) PRACTICAL **Marks: 60**

(i) Identification and mapping of relief, drainage pattern, water bodies and surface material from topographical maps

(ii) Identification and mapping of relief, drainage pattern, water bodies and surface material from aerial photographs and satellite imageries.

(iii) Identification and mapping of terrain units from large scale aerial photographs

(iv) Measurement and mapping of slopes from topographical maps

(v) Rough estimation and mapping of slopes from the aerial photographs

(vi) Morphometric analysis of drainage basins from topographical maps

(vii) Analysis of certain selective aspects of the morphometry of the drainage basins from the aerial photographs and satellite imageries

(viii) Empirical quantitative classification of terrain from topographical maps

(ix) Genetic and functional classification of terrain from the aerial photographs and satellite imageries

(x) Identification and mapping of impact of human activities on the geomorphological processes and landscape using remote sensing and GIS techniques

(xi) Identification of climate landscape relationships at micro-level from aerial photographs and satellite imageries

(xii) Management of natural environment pollution and waste using remote sensing and GIS techniques

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be based on class test/s and assignment/s etc., as prescribed or conducted by the teacher/s concerned.

BOOKS RECOMMENDED

ESSENTIAL READINGS

Agarwal, C.S. and P.K. Garg : *Text Book on Remote Sensing in Natural Resources Monitoring and Management*, Wheeler Publishing Co, New Delhi, 2000.

Druary, S.A. : *Image Interpretation in Geology*, Allen and Unwin Ltd., London, 1987.

Siegal, B.S. & Gillespie, A.R : *Remote Sensing in Geology*, John Wiley Publications, 1986.

Townsend, J.R.G. : *Terrain Analysis and Remote Sensing*, George Allen and Unwin, 1981.

Verstappen, H. : *Remote Sensing in Geomorphology*, Elseiver Scientific Publications, Netherlands, 1977.

Verbyla, David, L : *Satellite Remote Sensing of Natural Resources*, Lewis Publishers, New York, 2005

FURTHER READINGS

Greedry, Alan, F. : Application of Remote Sensing with Special Reference to Geosciences, Gregory Geo-science, 1974.

Nefedov, K.E. : Deciphering of Groundwater from Aerial Photographs, Amerind Publications, Delhi, 1972.

Papova, T.A. and Pandey, S.N.: Principles and Applications of Photogeology, Wiley Eastern Publishers, 1987

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Smith, Willam, L. : Remote Sensing Applications for Mineral Exploration, Dawden Hutchintons and Ross Inc., 1977.

PEDAGOGY:

- Students may be taken to institutions such as IIRS, NRSA, State Remote Sensing Cetres to acquaint them with equipments, techniques and their products.
- Students may be asked to prepare a report on landform using topographical sheets, aerial photographs and satellite images.
- Students may acquaint with the satellite imageries of various kinds of environmental problems

PAPER VI: IMAGE PROCESSING **Marks: 100**

(A) THEORY **Marks: 50**
Time : 3 Hours

OBJECTIVES:

The course is designed to help the students to understand image processing for image interpretation and pattern recognition. In addition, the techniques, which assist in providing focused and precise analysis of emerged pattern of geographical phenomena over the images following image processing.

Unit I

Image processing: definition and types-visual and digital
Visual image interpretation: Image characteristics, elements of object identification, image interpretation techniques, field verification.

Unit II

Digital Image Pre-processing: Sources of Errors, Types of error-Radiometric and Geometric, Image Rectification: Geometric Correction, Radiometric Correction, and Noise Removal.

Unit III

Image Enhancement Techniques: Contrast enhancement-Linear and Non-Linear, Logarithmic and Exponential, Gaussian stretch, Density slicing, Spatial Filtering: Low frequency and High frequency, Edge enhancement Band ratioing and band combination.

Unit IV

Digital image classification: Classification scheme-Supervised classification (training site selection and statistical information extraction), Discriminate functions: Maximum Likelihood classifier, Euclidian distance, Mahalanobis distance; Unsupervised Classification, Classification accuracy assessment, error matrix.

NOTE:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).

2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit, and the candidates will attempt 4 questions selecting one from each unit carrying 9 marks, in addition to the compulsory question at serial number I.

(B) PRACTICAL **Marks: 30**

(i) Familiarization with image processing system
(ii) Loading of image data, identification of objects on video display, study of histograms
(iii) Image enhancement techniques: contrast enhancement, band ratioing, edge enhancement, filtering, density slicing

(iv) Image registration: image to map, image to image
(v) Image classification techniques: supervised and unsupervised
(vi) Accuracy analysis
(vii) Ground data collection for training sets for classification of image

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 20**

Internal assessment shall be based on class test/s and assignment/s etc., as prescribed or conducted by the teacher/s concerned.

BOOKS RECOMMENDED

ESSENTIAL READINGS

Chanda B., Datta D, Majumdar : *Digital Image Processing and Analysis*, Prentice- Hall of India, 2001.

Fratt, W.K. : *Digital Image Processing*, John Wiley & Sons, 1978.

Jensen, J.R. : *Introductory Digital Image Processing: A Remote Sensing Perspective*, Prentice Hall, New Jersey, 1986.

Lillesand, T.M. & Kiefer, R.W. : *Remote Sensing and Image Interpretation*, Third Edition, John Wiley and Sons, 1994.

Prithvish Nag and M. Kudrat : *Digital Remote Sensing*, Concept Publishing Company, New Delhi, 1998.

FIELD REPORT **Marks: 100**

Distribution of Marks:

a) Evaluation of the Field Report : 50
b) Presentation & Viva Voce (25+25) : 50

NOTE:

The field report shall be prepared by the student on the basis of work done or experiences gained on visit(s) to Remote Sensing Centre(s)/Institute(s) in India. The student will submit three copies of his/her work ten days before the commencement of the theory examination of the said semester. Three examiners, consisting of supervisor, chairperson of the department, and one more faculty member teaching the course, shall evaluate the field report independently and student shall be awarded the mean score of the three evaluations. The evaluation process shall have two-components:(i) evaluation of field report, (ii) presentation of data analysis and main findings before the board of examiners and viva voce.

SEMESTER III

PAPER VII: APPLICATION OF REMOTE SENSING AND GIS IN DISASTER MANAGEMENT **Marks: 200**

(A) THEORY **Marks: 100**
Time : 3 Hours

OBJECTIVES:
Main objectives of this course are to introduce the students with the nature, causes and effects of disasters and applications of GIS and Remote Sensing in mapping, mitigation and monitoring of hazards and disasters.

Unit I

i. Fundamental concepts of hazards and disasters
ii. Natural and human induced disasters: classification, types
iii. Historical perspective of disasters in India

Unit II

iv. Disaster management: concepts relating to the pre-disaster phase, emergency phase, post-disaster phase
v. Disaster management mechanism in India: agencies, plans and policies
vi. Role of remote sensing and GIS in disaster management

Unit III

vii. Remote sensing and GIS application in geological hazards: Landslides and Earthquakes.
viii. Remote sensing and GIS application in Hydro meteorological Hazards: Floods, Cyclones, Drought.

Unit IV

ix. Remote sensing and GIS application in Environmental hazards: Deforestation, Land Degradation and desertification, Forest fires
x. Case Studies: Earthquakes in India, Drought in India, Floods in Indo-Gangetic plains, Landslides in Himalayan region,

Essential Readings

Greg Bankoff , Georg Frerks, : Dorothea Hilhorst — *Mapping Vulnerability: Disasters, Development and People*, Earthscan Publications Ltd., 2004.

Monmonier mark, : *Cartographies of Danger: Mapping Hazards in America***,** University of Chicago Press, 1997.

Roy, P.S.; Van Westen, C.J.; Jha, V.K.; Lakhera, R.C. and Champati Ray, P.K., : *Natural Disaster and their Mitigation: Remote Sensing and Geographical Information System Perspectives*, IIRS, DehraDun, Govt. of India, India, 2000.

FURTHER READINGS

Agarwal, C.S. and P.K. Garg : *Text Book on Remote Sensing in Natural Resources Monitoring and Management*, Wheeler Publishing Co, New Delhi, 2000.

Burrough, P.A. & Macdoneli, R.A.: *Principles of Geographic Information Systems,* Oxford University Press, 2000.

Demers, M.N. : *Fundamentals of geographic Information Systems*, 2nd Edition, John Wiley and sons, 2000.

Giarratano, J. & Rieley, G. : *Expert Systems: Principles and Programming*, PRS Kent Publishing Company, 1985.

Muralikrishna, I.V. : *Remote Sensing Applications and Geographic Information Systems Recent Trends*, Tata McGraw Hill, New Delhi, 1992.

Rashid, S.K. : *Remote Sensing in Geography*, Manak Publications, Delhi, 1993.

Rees, W.G. : *Physical Principles of Remote Sensing*, Cambridge University Press, Cambridge, 1990.

Swain, Phillip, H. : *Remote Sensing*: The quantitative Approach, McGraw Hill International Book Co., 1978.

Verbyla, David, L : *Satellite Remote Sensing of Natural Resources*, Lewis Publishers New York, 2005.

NOTE:

1. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt all 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit.
3. The candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(B) PRACTICAL **Marks: 60**

1. Earthquake hazard mapping using satellite images and ancillary data.
2. Flood prone area mapping using satellite images and ancillary data.
3. Landslide mapping and risk evaluation.
4. Multiple hazard mapping using satellite images, ancillary data and modeling risk in GIS.

(C) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be based on class test/s and assignment/s etc., as prescribed or conducted by the teacher/s concerned.

PEDAGOGY:

- Students may be asked to prepare a report on zonation of hazards and disasters using topographical sheets, aerial photographs and satellite images.
- Students may acquaint with the satellite imageries of various kinds of disasters.

PAPER VIII: APPLICATION OF REMOTE SENSING AND GIS IN UTILITY MANAGEMENT **Marks: 200**

(A) THEORY **Marks: 100**
Time : 3 Hours

OBJECTIVES:

The major objective of this paper is to expose the students the application of remote sensing and GIS in managing utility services. It further aims at training them to take quick, correct and reliable locational decisions.

Unit I

Utility, Description of all essential services and utilities, Database development and Data Acquisition, Acquiring and integrating geo-spatial data, Spatial Data Bases

Unit II

Spatial Data Manipulation and Analysis, Geospatial system analysis and design, Geospatial technology project, management, Query Processor and Visualization, Applications and Problem solving with GIS
Electricity, Gas, Water supply, Sewerage system
Solid waste disposal, Telecommunication, Public health and safety, Crime analysis

Unit III

Modelling in utility applications, Infrastructure aims and objectives, Environmental law and regulations governing infrastructure utilities, Modern infrastructure tools

Unit IV

Case studies Part I: relating to Electricity, Gas, Water supply, Sewerage system
Case studies Part II: Solid waste disposal, Telecommunication, Public health and safety, Crime analysis

REFERENCES

1. Harries K (1999) *Mapping Crime: Principle and Practice*. Washington, DC: Crime Mapping Research Center, Department of Justice, U.S.A.
2. Hodder I, Orton C (1979) *Spatial Analysis in Archaeology.* Cambridge: Cambridge University Press .

3. Escritt, L. B. (1972). *Water Supply and Building Sanitation*, 4th Ed., Mac Donald and Evans Limited, 1972
4. Hammer, Mark J. (1986). *Water and Waste Water Technology*, 2nd Ed., John Willey and Sons Inc., 1986.

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit, and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

(B) PRACTICAL **Marks: 60**

Site Suitability mapping and locating and relocating of infrastructure facilities related to:-

-Electricity and Gas,

-Water supply and Sewerage system,

-Telecommunications and Transport

-Site suitable for solid waste disposal, and

-Accessibility mapping of health care services.

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be based on class test/s and assignment/s etc., as prescribed or conducted by the teacher/s concerned.

Paper IX: APPLICATION OF REMOTE SENSING AND GIS IN RESOURCE MANAGEMENT **Marks: 200**

(A) THEORY **Marks: 100**
Time : 3 Hours

OBJECTIVES:

It aims at training the students to use remote sensing and GIS in management of resources. It will expose them to plan things better and utilize resources most efficiently.

Unit I

Concepts in Resources: Resources classification systems, natural and cultural resources, renewable and non-renewable resources.
Resource Conservation: Remote sensing based Land use- Land cover mapping for resource monitoring and management Sustainable development of natural resources.

Unit II

Land Resources: Introduction to agriculture and soil resources, Crop inventory, land evaluation, land degradation and soil conservation; Mineral resources and impacts of mining. Bio-Resources: Remote sensing application in agriculture, forest resources and wildlife habitat assessment. Mapping of forest density and type, issues in forest management.

Unit III

Water Resources: Remote sensing application in surface and sub surface water resources evaluation, watershed characterization, water resources management and development, water quality and pollution issues.
Energy Resources: Coal, oil and nuclear energy, non-conventional energy resources, future potential and requirement of energy resources. Geo-information in energy resources management.

Unit IV

Geo-information Models in Resource Management: Land resources development, prioritization and modeling, Forest resources modeling, Agriculture production potential modeling.

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit, and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

REFERENCES:

1. Simmons, T.G. (1974). Ecology of Natural Resources, Publ. Edward Arnold, London.
2. Chouhan, T.S. and Joshi, K.N. (1992). *Remote Sensing for Natural Resources Management*, Universal Scientific Publ. Jaipur.
3. Davidson, Donald A. (1998). *Soils and Land use Planning*, Longman, London.
4. Odum, E.P. (1964). *Fundamental of Ecology*, V.B. Saumders and Co.
5. Miller, R. W. and Donahue, R. L. (1990): *Soils*, Prentice-Hall of India.
6. Lillesand, Thomas M. & Kiefer Ralph (2000): *Remote Sensing and Image Interpretation*, John Wiley.
7. Colwell, Robert W. *Monitoring of Earth Resources from Aircraft and Spacecraft*, NASA, Washington D.C.
8. Ester, J. and Senger, L.W. *Remote Sensing Techniques for Environment Analysis*, Tayler and Francis Inc.

(B) PRACTICAL **Marks: 60**

1. LU-LC mapping at level I and Level II using 1:50,000 scale satellite image.
2. Forest Types Mapping using satellite images.
3. Delineating on satellite image various surface water resources and identify potential sites for WR conservation.
4. Delineation of surface mining zones and impact on land resources of the area using satellite image
5. NDVI and density slicing of digital satellite data for forest density classification.
6. Supervised classification for mapping agriculture and forest resources.
7. Applying geo-information models in resource management and development plan using raster GIS.
8. Soil erosion modeling using geo-information approach
9. Natural resource mapping and change detection study using temporal satellite data

NOTE: The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.

(C) INTERNAL ASSESSMENT **Marks: 40**

Internal assessment shall be based on class test/s and assignment/s etc., as prescribed or conducted by the teacher/s concerned.

SEMESTER IV

PAPER X: RESEARCH METHODOLOGY, PROJECT FORMULATION AND WRITING

Marks: 100
(Terminal exams: 80 marks)
(Internal Assessment: 20 marks)
Time : 3 Hours

OBJECTIVES:

The aim of this paper is to train the students about the various aspects of research writing so that they are in a position to write research reports independently.

Unit I

Problems of Geographical Information research: Identification of problems of regional and Local level, geographic data sources and nature of data to be used.
Hypotheses and Models
Formulation of research schemes

Unit II

Preparation of research projects and writing of reports: Preparation of field reports, spatial data, classification and sampling problems. Need for sampling, types of sampling, sample size, sampling area.
Project Definition, Importance of Projects and Project Management, Project Management context. Basics of project management, Project formulation, Time management, Budget estimates, Cost-benefit calculation techniques

Unit III

Project bidding, Project plan, Task definition, Project resource, Scheduling, Line management, Project team.
Managing the Projects Activities, Project administrator, Classification of projects. Product management, Problems and opportunities in Projects.

Unit IV

Tools & Methods: Project Communications and Presentation, Project Management Software, Project Administration.
Evolution, revolution, & termination of Project, Project change, End of projects, Project report preparation.

NOTE:

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit; and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

3. Mid-term appraisal shall include at least two presentations by the students before final submission of his/her project report. First presentation will be done after the collection of data/information and second one after the mapping and interpretation of the data/information. These presentations shall be assessed by a board of examiners consisting of Chairperson of the Department, faculty member teaching the course and one more senior faculty member.

REFERENCES:

1. W.E. Huxold & A.G. Lerinsons Aronoft. S. (1989). *Managing Geographic Information Projects.*

2. Earickson, R., and Harlin, J. (1994). *Geographic Measurement & Quantitative Analysis*, Macmillan, New York

3. Bennet P. Lientz & Kathryn P. (1995). *Project Management for the 21st Century*, Academic Press, California

PROJECT REPORT **Marks**: **500**

(a) Mid-Term Appraisal of the Project Report	:	100
(b) Evaluation of the Project Report	:	325
(c) Presentation &Viva Voce (50+25)	:	75

NOTE:

(i) Mid-term appraisal of the project report of each student shall be done by the board of the four examiners consisting of the supervisor, chairman, and the two senior faculty members of the department.

(ii) Mid-term appraisal of the project report shall have two-components: (i) presentation on project proposal including the research objectives, methodology and the chapter scheme, after the three weeks of assigning the project title, (ii) presentation of data analysis and tentative findings, after the six weeks of assigning the project title.

(iii) In case of project report, the Board of Control shall assign a candidate to a faculty member/s for supervision of his/her report on an approved topic. The student shall prepare three copies of his/her project report and submit in the concerned department. The report shall have to be submitted ten days before the commencement of the theory examination of the said semester. The project report shall incorporate the findings arrived at on the basis of the data/information collected from the field and processed, analyzed and mapped in the GIS lab.

For the evaluation of project report, three examiners, consisting of supervisor, chairperson of the department, and one more faculty member teaching the course, shall evaluate the field report, submitted by the student, independently and student shall be awarded the mean score of the three evaluations. The evaluation process shall have the two-components: (i) evaluation of report, (ii) presentation of data analysis and main findings before the board of examiners and viva voce.

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